List

List in python is a collection that store many items in one variable.

eg:shopping list

Square bracket []

mutuble - we can change after creation

allow duplicate values

ordered - it maintain the order

allow multiple datatype

String list

In [23]:

```
shop = input("enter list: ").split()
print(shop)
```

```
['Dress,Lipstick,Chain,Lipblam']
```

In [25]:

```
shop = input("Enter names: ").split(",")
print(shop)
```

```
['Dress', 'Lipstick', 'Chain', 'Lipblam']
```

Number List

In [36]:

```
num = list(map(int,input("enter num:").split(",")))
print(num)
```

```
[10, 30, 23]
```

In [38]:

```
num = list(map(int,input("enter num:").split()))
print(num)
```

```
[12, 45, 36]
```

1.list length

In [40]:

```
shop =["dress","bag","lipstick","chain"]
print(len(shop))
```

```
4
```

In [42]:

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```
mark=[45,76,89,54,67,77]
print(len(mark))
```

```
6
```

2.Type

In [55]:

```
#string
shop =["dress","bag","lipstick","chain"]
print(type(shop))
```

```
<class 'list'>
```

In [57]:

```
#int
mark=[45,76,89,54,67,77]
print(type(mark))
```

```
<class 'list'>
```

In [59]:

```
#float
per =[5.5,7.6,8.9,5.4,6.7,7.7]
print(type(mark))
```

```
<class 'list'>
```

In [61]:

```
#boolean
result = [True , True, False]
print(type(result))
```

```
<class 'list'>
```

3.access list

In [66]:

```
num =[10,5,3,6,8,2,3]
print(num[0:5])
```

```
[10, 5, 3, 6, 8]
```

In [68]:

```
num =[10,5,3,6,8,2,3]
print(num[0:])
```

```
[10, 5, 3, 6, 8, 2, 3]
```

In [70]:

```
num =[10,5,3,6,8,2,3]
print(num[:-1])
```

```
[10, 5, 3, 6, 8, 2]
```

In [72]:

```
num =[10,5,3,6,8,2,3]
print(num[::-1])
```

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```
[3, 2, 8, 6, 3, 5, 10]
```

In [74]:

```
num =[10,5,3,6,8,2,3]
print(num[3:5])
```

```
[6, 8]
```

In [76]:

```
num =[10,5,3,6,8,2,3]
print(num[0:7:2])
```

```
[10, 3, 8, 3]
```

4.changing list item

In [79]:

```
num =[10,5,3,6,8,2,3]
num[1]=20
print(num)
```

```
[10, 20, 3, 6, 8, 2, 3]
```

In [83]:

```
num =[10,5,3,6,8,2,3]
num[1:4]=20,30,40
print(num)
```

```
[10, 20, 30, 40, 8, 2, 3]
```

In [85]:

```
num =[10,5,3,6,8,2,3]
num[1:2]=20,30
print(num)
```

```
[10, 20, 30, 3, 6, 8, 2, 3]
```

5.insert item

In [89]:

```
num=[10,20,30,40]
num.insert(2,50)
print(num)
```

```
[10, 20, 50, 30, 40]
```

In [91]:

```
name = ["Reeta","Rithan","Kavi","Kannan"]
name.insert(2,"Prageesh")
print(name)
```

```
['Reeta', 'Rithan', 'Prageesh', 'Kavi', 'Kannan']
```

6.append

In [94]:

```
num=[10,20,30,40]
num.append(50)
print(num)
```

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```
[10, 20, 30, 40, 50]
```

In [96]:

```
name = ["Reeta","Rithan","Kavi","Kannan"]
name.append("Prageesh")
print(name)
```

```
['Reeta', 'Rithan', 'Kavi', 'Kannan', 'Prageesh']
```

7.extend

In [99]:

```
a=[10,20,30,40,50]
b=[60,70,80,90,100]
a.extend(b)
print(a)
```

```
[10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
```

In [101]:

```
a=[10,20,30,40,50]
b=[60,70,80,90,100]
a.append(b)
print(a)
```

```
[10, 20, 30, 40, 50, [60, 70, 80, 90, 100]]
```

8.remove spcific item

In [104]:

```
num = [10,15,20,30,40]
num.remove(15)
print(num)
```

```
[10, 20, 30, 40]
```

In [106]:

```
num =[10,15,20,30,15,40]
num.remove(15)
print(num)
```

```
[10, 20, 30, 15, 40]
```

9.pop()-remove specific item

In [109]:

```
a=[54,67,23,89,22,99]
a.pop(2)
print(a)
```

```
[54, 67, 89, 22, 99]
```

In [111]:

```
a=[54,67,23,89,22,99]
a.pop()
print(a)
```

```
[54, 67, 23, 89, 22]
```

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10.del

In [116]:

```
a=[1,2,3,4,5]
b=[6,7,8,9]
del b
print(a)
```

```
[1, 2, 3, 4, 5]
```

In [118]:

```
a=[1,2,3,4,5]
b=[6,7,8,9]
del b
print(b)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[118], line 4
      2 b=[6,7,8,9]
      3 del b
----> 4 print(b)

NameError: name 'b' is not defined
```

In [120]:

```
b=[6,7,8,9]
del b[3]
print(b)
```

```
[6, 7, 8]
```

11.Clear the list

In [128]:

```
a=[1,2,3,4,5]
a.clear()
print(a)
```

```
[]
```

12.sort list

In [133]:

```
a=[5,4,2,7,8,5]
a.sort()
print(a)
```

```
[2, 4, 5, 5, 7, 8]
```

In [135]:

```
a=[5,4,2,7,8,5]
a.sort(reverse=True)
print(a)
```

```
[8, 7, 5, 5, 4, 2]
```

In [137]:

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```
sub = ["Tamil","English","Maths","Science","History"]
sub.sort()
print(sub)
```

```
['English', 'History', 'Maths', 'Science', 'Tamil']
```

In [139]:

```
sub = ["tamil","english","maths","science","history"]
sub.sort()
print(sub)
```

```
['english', 'history', 'maths', 'science', 'tamil']
```

In [141]:

```
sub = ["Tamil","english","Maths","science","History"]
sub.sort()
print(sub)
```

```
['History', 'Maths', 'Tamil', 'english', 'science']
```

In [143]:

```
sub = ["Tamil","English","Maths","Science","History"]
sub.sort(reverse = True)
print(sub)
```

```
['Tamil', 'Science', 'Maths', 'History', 'English']
```

13.reverse order

In [146]:

```
a=[32,76,12,46,43]
a.reverse()
print(a)
```

```
[43, 46, 12, 76, 32]
```

14.copy a list

In [151]:

```
a=[1,2,3,4]
b=a.copy()
print(b)
```

```
[1, 2, 3, 4]
```

In [153]:

```
#another to copy
a=[1,2,3,4]
b=list(a)
print(b)
```

```
[1, 2, 3, 4]
```

In [155]:

```
# :slice also used to copy
a=[1,2,3,4]
b=a[:]
print(b)
```

```
[1, 2, 3, 4]
```

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15.Join a list

In [158]:
```
a=[1,2,3,4]
b=[34,67,34]
print(a+b)
```

```
[1, 2, 3, 4, 34, 67, 34]
```

In [160]:
```
a=[1,2,3,4]
b=[34,67,34]
a.extend(b)
print(a)
```

```
[1, 2, 3, 4, 34, 67, 34]
```

In [162]:
```
a=["apple","mango"]
a[:0] = ["kiwi","orange"]
print(a)
```

```
['kiwi', 'orange', 'apple', 'mango']
```

16.Nested if

In [169]:
```
a=[1,2,3,[4,5,[6,7],8,9],10,11]
print(a[1])
print(a[3][1])
print(a[3][2][0])
```

```
2
5
6
```

Practice Questions

1.Write a program to find the largest and smallest number in a list.

In [184]:
```
a=list(map(int,input("enter a:").split(",")))
largest=max(a)
smallest=min(a)
print("larger:",largest)
print("small:",smallest)
```

```
larger: 7
small: 1
```

2.Write a program to find the second largest number in a list.

In [199]:

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```
a=[2,5,4,7,6]
a.remove(max(a))
print(max(a))
```

```
6
```

3.Write a program to remove duplicates from a list.

In [204]:

```
a=[1,4,5,3,2,1,4,6,7,7]
print(list(set(a)))
```

```
[1, 2, 3, 4, 5, 6, 7]
```

4.Write a program to reverse a list without using reverse() or [::-1].

In [211]:

```
a=[3,6,2,8,9,1]
print(list(reversed(a)))
```

```
[1, 9, 8, 2, 6, 3]
```

5.Write a program to count how many times each element appears in a list.

In [214]:

```
a=[1,4,7,5,3,7,2,1,4,6,7,7]
appears=a.count(7)
print(appears)
```

```
4
```

6.Write a program to find the sum of all elements in a list.

In [227]:

```
a = [1, 2, 3, 4, 5]
print(sum(a))
```

```
15
```

7.Write a program to check if a list is a palindrome.

In [241]:

```
a = [1, 2, 3, 2, 1]
print(a == list(reversed(a)))
```

```
True
```

8.Write a program to merge two lists and sort the result.

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In [248]:

```
a=[12,3,4,22]
b=[54,67,23,11]
a.extend(b)
a.sort()
print(a)
```

```
[3, 4, 11, 12, 22, 23, 54, 67]
```

9.Write a program to find common elements between two lists.

In [251]:

```
a=[10,30,40,35]
b=[30,55,11,64]
common = set(a) & set(b)
print(common)
```

```
{30}
```

task

In [1]:

```
l=[5,36,"hello",7,32]
# 5,"hello","hi",64,32

l.remove(36)
l.insert(2,"hi")
l[3]=64
print(l)
```

```
[5, 'hello', 'hi', 64, 32]
```

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